2024, Mar. 2024 Release LOC:Acute Adult
Arrhythmia, Ventricular or Abnormal ECG Finding

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview

Instruction: This subset is for patients who present with ventricular arrhythmias and require evaluation and treatment. For atrial arrhythmias see the Arrhythmia: Atrial subset. For heart blocks see the Arrhythmia: Blocks subset. Criteria for the following ventricular arrhythmias are found within this subset:

- Wide complex tachycardia
- Accelerated idioventricular rhythm
- Ventricular tachycardia
- Ventricular fibrillation
- Torsades de pointes
- Abnormal electrocardiogram (ECG) (including: arrhythmogenic right ventricular cardiomyopathy, Wolff-Parkinson White, and Brugada syndrome)

Introduction:
Arrhythmias are often triggered by alcohol, myocardial infarction (MI), pericarditis, myocarditis, pulmonary embolism, pulmonary disease, hyperthyroidism, metabolic disease, electrolyte imbalance, or cardiac drug toxicity. Presenting symptoms of an arrhythmia often include: palpitations, chest pain, dyspnea, fatigue, lightheadedness, or syncope.
Evaluation/Intervention:
Patients presenting with an arrhythmia should undergo a thorough history and physical examination. A 12-lead ECG is recommended in all patients being evaluated for an arrhythmia. Additional testing may be necessary and includes: left ventricular functional assessment using echocardiogram, computed tomography (CT), cardiac magnetic resonance imaging (MRI), electrophysiology study (EPS), and laboratory testing.
Treatment goals for patients with an arrhythmia may include restoration of sinus rhythm, rate control, prevention of thromboembolism, and the prevention of future arrhythmias. Management may include discontinuation of proarrhythmic medication, initiation of antiarrhythmic medication, immunotherapy with digoxin specific antibody, insertion of an implantable cardioverter defibrillator (ICD) or pacemaker, ablation, or surgery (Zeppenfeld et al., Eur Heart J 2022, 43: 3997-4126; Littmann et al., The American journal of emergency medicine 2019:; Cronin et al., Heart rhythm 2019:epub; Brignole et al., Eur Heart J 2018, 39: 1883-948; Dan et al., Ep Europace 2018, 20: 731-2an).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:

- **Initial review, One:** [1]
  (From arrival in ED through decision to admit)
  (Symptom or finding within 24h)
  - **INTERMEDIATE, ≥ One:** [2]
    - Ventricular arrhythmia, ≥ **One:**
      - Accelerated idioventricular rhythm and systolic blood pressure ≥ 90 mmHg [3]
      - Ventricular tachycardia < 30 seconds [4]
    - Syncope or presyncope and, ≥ **One:** [5, 6]
      - Arrhythmogenic right ventricular cardiomyopathy (ARVC), actual or suspected [7]
      - Atypical Brugada pattern [8]
      - Brugada syndrome, actual or suspected [9, 10]
      - Long QT syndrome [11, 12]
      - Short QTc interval < 340 ms [13]
      - Wolff-Parkinson White (WPW), actual or suspected [14]
    - Sotalol initiation or dose adjustment (includes PO) ≤ 3d [15, 16]
  - **CRITICAL, ≥ One:** [17]
    - Accelerated idioventricular rhythm and systolic blood pressure < 90 mmHg [3]
    - ICD and shocks ≥ 3x/24h [18, 19]
    - Torsades de pointes [20]
    - Ventricular fibrillation [21, 22]
    - Ventricular tachycardia ≥ 30 seconds [4, 22]
    - Pre-hospital cardiac arrest and post resuscitation care ≤ 2d [23]
- **Episode Day 1, One:**
  (Symptom or finding within 24h)
  - **INTERMEDIATE, ≥ One:** [2]
    - Diagnosis, actual or suspected, **One:**
      - Acute coronary syndrome (use **Acute Coronary Syndrome** subset)
      - Fractured or displaced pacemaker lead (use **General Medical** subset)
      - Implantable device failure (use **General Medical** subset)
    - Ventricular arrhythmia, **Both:**
      - Finding, ≥ **One:**
        - Accelerated idioventricular rhythm and systolic blood pressure ≥ 90 mmHg [3]
        - Ventricular tachycardia < 30 seconds [4]
      - Intervention, ≥ **One:**
        - Antiarrhythmic [24, 25]
        - Atropine [26]
    - Syncope or presyncope and, **Both:** [5, 6]
      - Finding, ≥ **One:**
        - Arrhythmogenic right ventricular cardiomyopathy (ARVC), actual or suspected [7]
        - Atypical Brugada pattern [8]
        - Brugada syndrome, actual or suspected [9, 10]
        - Long QTc interval [11, 12]
        - Short QTc interval < 340 ms [13]
        - Wolff-Parkinson White (WPW), actual or suspected [14]
      - Intervention, ≥ **One:**
        - Antiarrhythmic (includes PO) [27]
        - Imaging or evaluation, ≥ **One:**
          - Cardiac MRI scheduled [28]
          - Electrophysiology study (EPS) evaluation ≤ 24h [29, 30]
          - ICD placement or evaluation ≤ 24h [10]
          - Ablation evaluation ≤ 24h [31]

- Sotalol initiation or dose adjustment (includes PO) ≤ 3d [15, 16]
- **CRITICAL,** ≥ **One:** [17]
  - Arrhythmia, **Both:**
    - Finding, ≥ **One:**
      - Accelerated idioventricular rhythm and systolic blood pressure < 90 mmHg [3]
      - ICD and shocks ≥ 3x/24h [18, 19]
      - Torsades de pointes [20]
      - Ventricular fibrillation [21, 22]
      - Ventricular tachycardia ≥ 30 seconds [4, 22]
    - Intervention, ≥ **One:**
      - Antiarrhythmic
      - Atropine [26]
      - Digoxin toxicity and administration of digoxin-specific antibody [32]
      - Cardioversion, electrical performed within last 24h [33]
      - Defibrillation performed within last 24h
      - ICD placement or evaluation ≤ 24h [10]
      - Temporary pacemaker ≤ 2d [34]
      - Torsades de pointes and magnesium sulfate [20, 35]
  - Pre-hospital cardiac arrest and post resuscitation care ≤ 2d [23]
  - CRRT [36]
  - ECMO or ECLS [37]
  - Hemodynamic monitoring, invasive [38]
  - IABP [39]
  - Mechanical ventilation
  - NIPPV and, ≥ **One:** [40]
    - Arterial Po_2 ≤ 39 mmHg(5.2 kPa)
    - Arterial or venous Pco_2 ≥ 60 mmHg(8.0 kPa) and, **One:**
      - COPD and arterial or venous pH ≤ 7.24
      - No hx of COPD
  - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 2, One:**
  - **ACUTE, One:** [41]
    - Acute coronary syndrome (use **Acute Coronary Syndrome** subset)
    - Antiarrhythmic medication adjustment (includes PO) ≤ 2d [42]
  - **INTERMEDIATE,** ≥ **One:**
    - Acute coronary syndrome (use **Acute Coronary Syndrome** subset)
    - Arrhythmia, ≥ **One:**
      - IV antiarrhythmic and hemodynamically stable
      - Antiarrhythmic conversion from IV to PO, **One:**
        - No proarrhythmic risk ≤ 24h
        - Proarrhythmic risk and continuous cardiac monitoring (excludes Holter) ≤ 3d [43]
    - Arrhythmogenic right ventricular cardiomyopathy (ARVC) confirmed and, ≥ **One:**
      - ICD placement or evaluation ≤ 24h
      - Initiation of sotalol (includes PO) ≤ 3d [16, 44]
    - Wolff-Parkinson White (WPW) confirmed and, **One:** [14]
      - Ablation ≤ 24h [27]
      - Initiation of sotalol (includes PO) ≤ 3d [16, 44, 45]
    - ICD placement ≤ 24h [18]
    - NIPPV [40]
    - Oxygen ≥ 40%(0.40), ≤ 2d [46]
    - Permanent pacemaker placement ≤ 24h
    - Post cardioversion ≤ 24h
    - Post Critical care ≤ 24h

- Sotalol initiation or dose adjustment (includes PO) ≤ 3d [15, 16]
- **CRITICAL, ≥ One:**
  - Acute coronary syndrome (use **Acute Coronary Syndrome** subset)
  - Antiarrhythmic and hemodynamic instability and, **Both:**
    - Finding, ≥ **One:**
      - Systolic blood pressure, ≥ **One:**
        - < 90 mmHg
        - < 110 mmHg with chronic hypertension
        - > 30 mmHg decrease from baseline [47]
        - Labile [48]
      - MAP < 65 mmHg
    - Cardioversion, electrical performed ≤ 24h [49]
  - Pre-hospital cardiac arrest and post resuscitation care ≤ 2d [23]
  - CRRT [36]
  - Defibrillation performed within last 24h
  - ECMO or ECLS [37]
  - Hemodynamic monitoring, invasive [38]
  - IABP [39]
  - Mechanical ventilation
  - NIPPV and, ≥ **One:** [40]
    - FiO_2 > 50%(0.50)
    - Respiratory intervention every 1-2h [50]
  - Temporary pacemaker insertion performed ≤ 2d
  - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 3, One:**
  - **ACUTE, One:**
    - **Responder,** discharge expected today if clinically stable last 12h, **All:** [51]
      - Systolic blood pressure within acceptable limits or baseline [52]
      - Arrhythmia controlled or resolved
      - Medication regimen established and tolerated
      - Complication or comorbidity, **One:**
        - No complication or active comorbidity relevant to this episode of care [53]
        - Heart failure resolving or at baseline [47]
    - **Partial Responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
      - Antiarrhythmic medication adjustment (includes PO) ≤ 2d [42]
      - Heart failure and, **Both:** [54]
        - Symptoms or findings, ≥ **One:** [47]
          - O_2 sat ≤ 91%(0.91) and < baseline [47]
          - Worsening edema
          - Heart failure confirmed by imaging [55]
        - Intervention, ≥ **One:** [56]
          - Diuretic, **One:** [57]
            - ≥ 2x/24h (includes PO)
            - Requires daily medication adjustment (includes PO) and BUN or creatinine increased from baseline [42, 47]
          - Dialysis
      - Post Critical care ≤ 24h
  - **INTERMEDIATE, One:**
    - **Responder,** discharge expected today if clinically stable last 12h, **All:** [51]
      - Arrhythmia controlled or resolved
      - Medication regimen established and tolerated
      - Systolic blood pressure within acceptable limits or baseline [52]
      - O_2 sat within acceptable limits [58]

- **Partial Responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
  - Arrhythmia, ≥ **One:**
    - IV antiarrhythmic and hemodynamically stable
    - Antiarrhythmic conversion from IV to PO, **One:**
      - No proarrhythmic risk ≤ 24h
      - Proarrhythmic risk and continuous cardiac monitoring (excludes Holter) ≤ 3d [43]
  - ICD placement ≤ 24h [18]
  - NIPPV [40]
  - Oxygen ≥ 40%(0.40), ≤ 2d [46]
  - Permanent pacemaker ≤ 24h
  - Post cardioversion ≤ 24h
  - Post Critical care ≤ 24h
  - Sotalol initiation or dose adjustment (includes PO) ≤ 3d [15, 16]
- **CRITICAL, ≥ One:**
  - Antiarrhythmic and hemodynamic instability and, **Both:**
    - Finding, ≥ **One:**
      - Systolic blood pressure, ≥ **One:**
        - < 90 mmHg
        - < 110 mmHg with chronic hypertension
        - > 30 mmHg decrease from baseline [47]
        - Labile [48]
      - MAP < 65 mmHg
    - Cardioversion, electrical performed within last 24h [49]
  - CRRT [36]
  - Defibrillation performed within last 24h
  - ECMO or ECLS [37]
  - Hemodynamic monitoring, invasive [38]
  - IABP [39]
  - Mechanical ventilation
  - NIPPV and, ≥ **One:** [40]
    - FiO_2 > 50%(0.50)
    - Respiratory intervention every 1-2h [50]
  - Temporary pacemaker insertion performed within last 2d
  - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

- **Episode Day 4, One:**
  - **ACUTE, One:**
    - **Responder,** discharge expected today if clinically stable last 12h, **All:** [51]
      - Systolic blood pressure within acceptable limits or baseline [52]
      - Arrhythmia controlled or resolved
      - Medication regimen established and tolerated
      - Complication or comorbidity, **One:**
        - No complication or active comorbidity relevant to this episode of care [53]
        - Heart failure resolving or at baseline [47]
    - **Partial Responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
      - Antiarrhythmic medication adjustment (includes PO) ≤ 2d [42]
      - Heart failure and, **Both:** [54]
        - Symptoms or findings, ≥ **One:** [47]
          - O_2 sat ≤ 91%(0.91) and < baseline [47]
          - Worsening edema
          - Heart failure confirmed by imaging [55]
        - Intervention, ≥ **One:** [56]
          - Diuretic, **One:** [57]
            - ≥ 2x/24h (includes PO)

- Requires daily medication adjustment (includes PO) and BUN or creatinine increased from baseline [42, 47]
- Dialysis
- Post Critical care ≤ 24h

- **INTERMEDIATE, One:**
  - **Responder,** discharge expected today if clinically stable last 12h, **All:** [51]
    - Arrhythmia controlled or resolved
    - Medication regimen established and tolerated
    - Systolic blood pressure within acceptable limits or baseline [52]
    - O_2 sat within acceptable limits [58]
  - **Partial Responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
    - Arrhythmia, ≥ **One:**
      - IV antiarrhythmic and hemodynamically stable
      - Antiarrhythmic conversion from IV to PO, **One:**
        - No proarrhythmic risk ≤ 24h
        - Proarrhythmic risk and continuous cardiac monitoring (excludes Holter) ≤ 3d [43]
    - ICD placement ≤ 24h [18]
    - NIPPV [40]
    - Oxygen ≥ 40%(0.40), ≤ 2d [46]
    - Permanent pacemaker ≤ 24h
    - Post cardioversion ≤ 24h
    - Post Critical care ≤ 24h
- **CRITICAL,** ≥ **One:**
  - Antiarrhythmic and hemodynamic instability and, **Both:**
    - Finding, ≥ **One:**
      - Systolic blood pressure, ≥ **One:**
        - < 90 mmHg
        - < 110 mmHg with chronic hypertension
        - > 30 mmHg decrease from baseline [47]
        - Labile [48]
      - MAP < 65 mmHg
    - Cardioversion, electrical performed within last 24h [49]
  - CRRT [36]
  - Defibrillation performed within last 24h
  - ECMO or ECLS [37]
  - Hemodynamic monitoring, invasive [38]
  - IABP [39]
  - Mechanical ventilation
  - NIPPV and, ≥ **One:** [40]
    - FiO_2 > 50%(0.50)
    - Respiratory intervention every 1-2h [50]
  - Temporary pacemaker insertion performed within last 2d
  - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

Episode Day 5, One:

- **ACUTE, One:**
  - **Responder,** discharge expected today if clinically stable last 12h, **All:** [51]
    - Systolic blood pressure within acceptable limits or baseline [52]
    - Arrhythmia controlled or resolved
    - Medication regimen established and tolerated
    - Complication or comorbidity, **One:**
      - No complication or active comorbidity relevant to this episode of care [53]
      - Heart failure resolving or at baseline [47]
  - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for

this condition, **One:**
- **Use Extended Stay subset**
- Refer for secondary review

- **INTERMEDIATE, One:**
  - **Responder,** discharge expected today if clinically stable last 12h, **All:** [51]
    - Arrhythmia controlled or resolved
    - Medication regimen established and tolerated
    - Systolic blood pressure within acceptable limits or baseline [52]
    - O_2 sat within acceptable limits [58]
  - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**
    - **Use Extended Stay subset**
    - Refer for secondary review
- **CRITICAL, One:**
  - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
    - **Use Extended Stay subset**
    - Refer for secondary review

Discharge, Level of Care, One: [59]
- **HOME, Both:**
  - Level of care appropriateness, **Both:**
    - Home environment safe and accessible [60]
    - Patient or caregiver demonstrates ability to manage care
  - Arrhythmia discharge planning, **All:**
    - Education on the importance of, **All:**
      - Strict medication adherence
      - Participating in an exercise program
      - Avoiding tobacco, excessive alcohol, and illicit drug use
      - Understanding when and where to seek help
    - Provide comprehensive written discharge and teaching instructions [61]
    - Medication reconciliation [62]
    - Follow-up care planned [63]
    - Identify and address transportation needs
- **HOME CARE, Both:**
  - Level of care appropriateness, **All:**
    - Home environment safe and accessible [60]
    - Patient or caregiver willing and able to learn care
    - Outpatient management contraindicated or unavailable [64, 65]
    - Skilled services, ≥ **One:** [66]
      - Skilled nursing [67]
      - Skilled therapy, PT, OT, or ST
      - Behavioral health [68]
  - Arrhythmia discharge planning, **All:**
    - Education on the importance of, **All:**
      - Strict medication adherence
      - Participating in an exercise program
      - Avoiding tobacco, excessive alcohol, and illicit drug use
      - Understanding when and where to seek help
    - Provide comprehensive written discharge and teaching instructions [61]
    - Medication reconciliation [62]
    - Follow-up care planned [63]
    - Identify and address transportation needs

- **BEHAVIORAL HEALTH, Both:**
  - Level of care appropriateness, **One:**
    - Support systems available [69]
  - Skilled treatment, ≥ **One:**
    - Clinical assessment [70]
    - Individual, group, or family counseling
    - Psychiatric, substance, or medication evaluation [71]
- **SKILLED MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
    - Treatment precluded in a lower level
    - Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 1-2x/24h [67]
      - Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [72, 73]
        - Functional impairments requiring at least supervision [74]
        - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [62]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
    - Treatment precluded in a lower level
    - Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 4h/24h [67]
      - Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [72, 73]
        - ≥ 2 functional impairments requiring at least minimum assistance [74]
        - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [62]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **SUBACUTE REHAB, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Treatment precluded in a lower level
    - Services, **All:**
      - Goal directed therapy with rehabilitation potential [72, 73]
      - ≥ 2 functional impairments requiring at least minimum assistance [74]
      - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [62]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **ACUTE REHAB, Both:**
  - Level of care appropriateness, **Both:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk

- Services, **All:**
  - Goal directed therapy with rehabilitation potential [72, 73]
  - ≥ 2 functional impairments requiring at least minimum assistance [74]
  - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines [75]
- Complete prior to facility transfer, **All:**
  - Medication reconciliation [62]
  - Obtain and complete forms for facility
  - Obtain discharge summary and transmit to facility and medical practitioner
  - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
  - Level of care appropriateness, **Both:**
    - Treatment precluded in a lower level
    - In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One:**
      - Primary condition or illness and treatment of active comorbid condition(s) [76]
      - Ventilator dependent ≥ 6h/d and weaning planned [77, 78]
      - Acute and comorbid conditions requiring prolonged hospitalization [79]
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [62]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
308 CARDIAC ARRHYTHMIA AND CONDUCTION DISORDERS WITH MCC	**n/a**	**3.5**	**CMS GMLOS**
309 CARDIAC ARRHYTHMIA AND CONDUCTION DISORDERS WITH CC	**n/a**	**2.3**	**CMS GMLOS**
310 CARDIAC ARRHYTHMIA AND CONDUCTION DISORDERS WITHOUT CC/MCC	**n/a**	**1.8**	**CMS GMLOS**
VENTRICULAR FIBRILLATION	**6%**	**5.9**	**InterQual**
VENTRICULAR TACHYCARDIA	**n/a**	**3.6**	**InterQual**

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:

1:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

2:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

3:

An accelerated idioventricular rhythm (AIVR) is an ectopic ventricular rhythm characterized by three or more consecutive PVCs occurring at a rate faster than the normal ventricular escape rate of 30 to 40 beats per minute. It may occur with an MI, drug toxicity, or inflammatory processes. Similar to ventricular tachycardia, AIVR can be differentiated by a slower rate. In general, the upper rate limit for AIVR is 100 to 120 beats per minute. In patients with left ventricular dysfunction who do not tolerate AIVR because of the loss of atrioventricular synchrony, acute treatment includes increasing the atrial rate with intravenous atropine or a temporary pacemaker.

4:

Ventricular tachycardia (VT) is a repetitive, electrical stimulation of the ventricles resulting in an uncontrolled rapid heartbeat with loss of effective ventricular contractility which may lead to ventricular fibrillation. VT is classified as non-sustained when it lasts less than 30 seconds and sustained when it lasts longer than 30 seconds or requires intervention due to hemodynamic compromise (Zeppenfeld et al., Eur Heart J 2022, 43: 3997-4126).

5:

Syncope is the transient loss of consciousness and postural tone caused by diminished cerebral blood flow characterized by rapid onset, short duration, and spontaneous complete recovery.

6:

Patients with a known history of arrhythmogenic syncope are 50% to 75% more likely to have future arrhythmic events compared to asymptomatic patients. Arrhythmias that are likely to cause a syncopal episode include (Brignole et al., Eur Heart J 2018, 39: 1883-948):

- Consistent bradycardia < 40 beats per minute or sinus pauses > 3 seconds while awake (without physical training)
- Mobitz II- 2nd and 3rd degree atrioventricular (AV) block
- Alternating left and right bundle branch block (BBB)
- Ventricular tachycardia (VT) or rapid paroxysmal supraventricular tachycardia (SVT)
- Non-sustained episodes of polymorphic VT and long or short QT interval
- Pacemaker or implantable cardioverter defibrillator (ICD) failure with cardiac pauses

7:

Electrocardiogram (ECG) findings suggestive of arrhythmogenic right ventricular cardiomyopathy include inverted T waves in the right precordial leads, epsilon waves, and ventricular late potentials. An epsilon wave is described as a terminal notch in the QRS complex and can be caused by slowed intraventricular conduction. Ventricular late potentials are small-amplitude, short-duration waves that occur after the QRS complex and can be precursors to cardiac arrhythmias (Zeppenfeld et al., Eur Heart J 2022, 43: 3997-4126; Brignole et al., Eur Heart J 2018, 39: 1883-948).

8:

An electrocardiogram (ECG) finding of a J wave, ST-segment elevation in the right precordial leads, and the inferior or high lateral leads is consistent with atypical Brugada pattern.

9:

A right bundle branch block with ST elevation in leads V_1, V_2, and V_3 is suggestive of Brugada pattern.

10:

An implantable cardioverter defibrillator (ICD) is the preferred treatment for Brugada syndrome with syncope because of the high risk of subsequent arrhythmic events (sustained ventricular tachycardia, ventricular fibrillation, or sudden death). A family history of sudden death, positive electrophysiology study (EPS), fractionated QRS, early repolarization in the peripheral leads, increased T-peak-to-T-end interval, and long PR interval are risk factors which favor ICD placement in Brugada syndrome (Brignole et al., Eur Heart J 2018, 39: 1883-948; Wu et al., Medicine (Baltimore) 2016, 95: e4214).

11:

Long QT syndrome is diagnosed with a QTc interval ≥ 500 ms. Long QT syndrome is also diagnosed with a QTc interval of ≥ 480 to 499 ms when associated with syncope (Shen et al., Journal of the American College of Cardiology 2017, 70: e39-e110).

12:

Guidelines recommend beta-blockers as the first-line therapy for long QT syndrome. If a patient continues to be symptomatic despite beta-blocker therapy or is non-compliant they should undergo an ICD placement (Shen et al., Journal of the American College of Cardiology 2017, 70: e39-e110).

13:

Short-QT syndrome is a rare, genetic disease with the clinical finding of the QT interval less than 340 ms. Implantable cardioverter defibrillator (ICD) placement may be of benefit for patients with a history of ventricular arrhythmias or a family history of sudden cardiac death (Shen et al., J Am Coll Cardiol 2017, 70: e39-e110).

14:

Wolff-Parkinson White (WPW) is one of the most common pre-excitation syndromes. This occurs when the ventricle is activated early by an abnormal pathway which bypasses the normal conduction system. The classic WPW electrocardiogram pattern is a short PR interval and ventricular pre-excitation leading to tachyarrhythmia.

15:

Instruction: Application of this criteria point applies to patients who require hospitalization for the purpose of cardiac monitoring while sotalol dose is being initiated or adjusted. Sotalol may prolong the QT interval potentially worsening an existing arrhythmia or precipitating a new one.

16:

Initial refers to the first time a medication or treatment is utilized. During the period of initiation, if the medication or treatment is temporarily discontinued up to 24 hours (therapeutic pause) it is still considered initial. If a tolerated medication or treatment is discontinued for more than 24 hours and then restarted, it is not considered initial. Medications that were initiated on an outpatient basis and are discontinued prior to hospitalization may be considered initial based on the need for monitoring or duration of discontinuation. Medications that were temporarily held due to concurrent illness and are resumed are not considered initial.

17:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

18:

An Implantable Cardioverter Defibrillator (ICD) initiates an electrical stimulation to the heart after detecting ventricular fibrillation and restores the normal heart rhythm.

19:

Instruction: Patients with an implantable cardioverter defibrillator (ICD) experiencing multiple shocks may be experiencing failure of their device or worsening cardiac pathology. Hospital admission for device interrogation, antiarrhythmic medication adjustment, or evaluation is often indicated. Inappropriate shocks are more common in patients with atrial fibrillation and are associated with a higher mortality.

20:

Torsades de pointes is a ventricular tachycardia that displays a gradual alteration in the amplitude and direction of electrical activity and occurs with a prolonged QT interval. It should be considered when ventricular tachycardia is unresponsive to treatment. The most common causes include drugs (e.g., Class I and III antiarrhythmics, antihistamines), electrolytes imbalances (e.g., hypokalemia, hypomagnesemia), myocardial ischemia, and central nervous system disorders (Zeppenfeld et al., Eur Heart J 2022, 43: 3997-4126).

21:

Ventricular fibrillation is a rapid, grossly irregular ventricular rhythm characterized by marked variability in the QRS cycle length, morphology, and amplitude (Zeppenfeld et al., Eur Heart J 2022, 43: 3997-4126).

22:

Patients with cardiac arrest related to sustained ventricular tachycardia or fibrillation should be evaluated and treated for myocardial ischemia.

23:

Mortality post cardiac arrest is high with most deaths occurring within the first 24 hours. Systematic post-cardiac arrest care after the return of spontaneous circulation can improve the likelihood of survival with good quality of life. Multiple organ systems are affected after cardiac arrest, thus development of a system wide proactive plan of care may be beneficial for cardiac arrest survivors.
In adult patients, post resuscitation care should include (Callaway et al., circulation 2015, 132: S465-S82):

- Therapeutic hypothermia for those patients unable to follow commands after resuscitation
- Optimization of hemodynamics and gas exchange
- Immediate coronary reperfusion with percutaneous coronary intervention (PCI) when indicated
- Glycemic control
- Neurological management and prognostication

In pediatric patients, post resuscitation care should include (Chameides et al., Pediatric Advanced Life Support: Provider Manual Pap/Crds P ed. 2012):

- Maintaining cardiovascular function and promoting adequate tissue perfusion
- Optimizing oxygenation and ventilation
- Glycemic control
- Correction of electrolyte or acid-base abnormalities
- Maintaining appropriate levels of analgesia and sedation

- Consideration of therapeutic hypothermia if coma persists following resuscitation

24:
Initial treatment for ventricular arrhythmias frequently includes antiarrhythmics (Zeppenfeld et al., Eur Heart J 2022, 43: 3997-4126).

25:
Hemodynamic stability is evidenced by adequate tissue and organ perfusion. Blood pressure and heart rate are parameters commonly used to assess hemodynamic stability. Clinical findings associated with hemodynamic stability include normal peripheral pulses, warm extremities, adequate urine output, and neurological stability. The Critical level of care is appropriate for patients who develop hemodynamic instability.

26:
Atropine can increase sinus node firing in symptomatic accelerated idioventricular rhythm (AIVR).

27:
The most serious complication in patients with Wolff-Parkinson-White (WPW) syndrome is atrial fibrillation (AF) causing rapid conduction of atrial impulses across the accessory pathway resulting in ventricular fibrillation and sudden cardiac death. For patients with WPW or pre-excitation syndrome who have AF or atrioventricular reentrant tachycardia, catheter ablation should be considered as first-line therapy (Kirchhof et al., Europace 2016, 18: 1609-78; Page et al., J Am Coll Cardiol 2016, 67: e27-e115; January et al., J Am Coll Cardiol 2014, 64: e1-76).

28:
Cardiac MRI is recommended if arrhythmogenic right ventricular cardiomyopathy (ARVC) is suspected (Brignole et al., Eur Heart J 2018, 39: 1883-948).

29:
An electrophysiological study (EPS) involves the introduction of multipolar catheter electrodes into the venous or arterial system to record and stimulate cardiac electrical activity. It is used to terminate a tachycardia by electrical stimulation or shock, to evaluate how an intervention modifies or prevents tachyarrhythmias, to assess the characteristics of an arrhythmia before device implantation, to test the function of an implanted device, or to identify patients at risk for sudden cardiac death.

30:
An electrophysiology study (EPS) can be useful to diagnose arrhythmogenic right ventricular cardiomyopathy (ARVC). Ventricular tachycardia (VT) induction during an EPS is suggestive of ARVC (Brignole et al., Eur Heart J 2018, 39: 1883-948).

31:
Reference: (Cronin et al., Heart rhythm 2019:epub)

32:
For hemodynamically unstable patients, or those with lethal arrhythmia, immunotherapy with digoxin-specific antibodies (known as Fab fragments) may be administrated via an intravenous (IV) bolus to bind intravascular free digoxin. A complete response usually occurs within 4 hours. In patients with heart failure, digoxin-specific antibodies should be used with caution. Patients should be monitored for signs of volume overload and hypokalemia due to movement of intravascular potassium into the cell (Kusumoto et al., Circulation 2019, 140: e382-e482).

33:
Electrical cardioversion for atrial fibrillation (AF) should only be performed if the confirmed onset is within 48 hours or there is no prior history of stroke or transient ischemic attack (TIA) within 6 months. Patients with a history of valvular disease are contraindicated for electrical cardioversion, and those with electrolyte abnormalities

or digoxin toxicity should be managed with rate control only (Atzema and Singh, Cardiol Clin 2018, 36: 141-59). Electrical cardioversion is also indicated in hemodynamically unstable patients caused by AF or those that are hemodynamically stable but failed pharmacological conversion or performed as the initial therapy for rhythm control (Joglar et al., J Am Coll Cardiol 2023:).

34:

Temporary pacemaker placement is recommended for patients with symptomatic bradycardia or heart block (Kusumoto et al., Circulation 2019, 140: e382-e482; Oken et al., JAMA cardiology 2019:).

35:

Intravenous (IV) magnesium sulfate replacement is indicated in patients with long QT syndrome and few episodes of Torsades de pointes. Treatment with magnesium sulfate is not likely to be effective in patients with a normal QT interval (Baker, European Heart Journal-Cardiovascular Pharmacotherapy 2016, 3: 108-17).

36:

While no randomized, controlled trials have shown that continuous renal replacement therapy (CRRT) is superior to intermittent dialysis with respect to survival, there are several advantages that may influence its use despite the lack of demonstrated survival benefit.
These advantages include:

- Gradual solute and fluid shifts resulting in greater hemodynamic stability, little risk of disequilibrium syndrome, and no worsening of brain edema
- Less need for fluid or nutritional restrictions, allowing for adequate nutrition without compromising fluid balance
- Decreased risk for intradialytic hypotension and recurrent kidney injury

Comprehensive guidelines recommend the use of CRRT for patients with hemodynamic instability, brain edema, persistent metabolic acidosis, and large fluid removal requirements. CRRT can be discontinued once renal recovery has been confirmed or another form of renal replacement is chosen based on the patient's clinical condition (Ostermann et al., Kidney Int 2020, 98: 294-309; Depret et al., Ann Intensive Care 2019, 9: 32; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138).

37:

Extracorporeal membrane oxygenation (ECMO) or extracorporeal life support (ECLS) provides oxygen and removes carbon dioxide from blood. ECMO/ECLS facilitates the recovery of cardiopulmonary function when other measures fail to improve gas exchange. Indications for the use of ECMO/ECLS in children include refractory septic or cardiogenic shock, cardiac arrest as a bridge to transplant, or refractory hypoxemic or hypercarbic respiratory failure. There is clear evidence to support the use of ECMO/ECLS in the pediatric population, but definitive studies in adults are lacking. However, in adults, observational studies and other data support the use of ECMO/ECLS for severe refractory, hypoxemic, or hypercarbic respiratory failure (Mehta et al., American College of Cardiology Perspectives and Analysis 2018; Napp et al., Herz 2017, 42: 27-44; Guirand et al., J Trauma Acute Care Surg 2014, 76: 1275-81). ECMO/ECLS incorporates the use of an external circuit, a cannula for vascular access, a pump to propel the blood through the circuit, and an artificial membrane that oxygenates the blood. Common complications include hemorrhage, hemolysis, and infection. Stroke, embolism, and seizures in children have also been reported. Infants with intraventricular hemorrhage (IVH) or those at high risk for IVH (i.e., weight less than 2000 grams, age less than 35 weeks) are not candidates for this therapy (Abrams et al., J Am Coll Cardiol 2014, 63: 2769-78; Domico et al., Pediatr Crit Care Med 2012, 13: 16-21; MacLaren et al., Intensive Care Med 2012, 38: 210-20; Peek et al., Health Technol Assess 2010, 14: 1-46; Mugford et al., Cochrane Database Syst Rev 2008: CD001340).

38:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

39:

An intra-aortic balloon pump (IABP) is a mechanical circulatory support device that utilizes a catheter-mounted intravascular device positioned in the descending thoracic aorta. The balloon automatically inflates during early diastole, increasing coronary blood flow and deflates during early systole, reducing afterload as the left ventricle ejects. Indications for use include cardiogenic shock, temporary support of left ventricular function after myocardial infarction (MI), preservation of myocardial viability, or refractory ventricular arrhythmias post-MI or preoperative bridge before coronary artery bypass graft (CABG) surgery. Contraindications include aortic regurgitation and aortic dissection.

40:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).
The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.
Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

41:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

42:

Instruction: For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

43:

Instruction: Examples of antiarrhythmics with proarrhythmic risk include, but are not limited to, amiodarone, disopyramide, dofetilide, dronedarone, flecainide, mexiletine, propafenone, and sotalol. Many factors influence the appropriate choice of drug, including underlying medical conditions and concurrent drug administration. In some instances, such as patients who are pregnant or are breastfeeding, there are drugs which are contraindicated or have limited evidence to support their use.

44:

Cardioversion is used to restore sinus rhythm and can be either chemical (with medications) or electrical. While the use of direct-current cardioversion is more effective, it also requires conscious sedation or anesthesia. The development of new antiarrhythmics has made chemical cardioversion a more commonly used method. Risks associated with chemical cardioversion include drug-induced torsades de pointes or other serious arrhythmias. This risk can be mitigated by monitoring the patient during initiation of an antiarrhythmic. Choice of cardioversion method is determined by several factors including patient stability and treatment history. Cardioversion carries the risk of thromboembolism, regardless of the method, unless anticoagulation has been previously initiated. Risk of thromboembolism is highest when the arrhythmia has been present for longer than 48 hours.

45:

Drugs with a high proarrhythmic risk may cause worsening of an existing arrhythmia or precipitate a new one. Cardiac monitoring is recommended during initiation of these medications (January et al., Heart Rhythm 2019, 16: e66-e93).

46:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO_2) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO_2 with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO_2 retention, where a precise inspired FiO_2 is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO_2. The following are estimates of O_2 delivered at the associated flow rates:

LOW-FLOW SYSTEMS
Nasal cannula
Room air 21%(*0.21*)
1L 24%(*0.24*) **4L** 36%(*0.36*)
2L 28%(*0.28*) **5L** 40%(*0.40*)
3L 32%(*0.32*) **6L** 44%(*0.44*)
Simple oxygen masks

- 35-50%(*0.35-0.50*) FiO_2 (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO_2 in the mask

Venturi masks

- 24-50%(*0.24-0.50*) FiO_2 (4-12 L/min)

Partial rebreathing masks

- 40-70%(*0.40-0.70*) FiO_2 (6-10 L/min)

Non-rebreathing masks

- 60-80%(*0.60-0.80*) FiO_2 (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS
Air-entrainment masks/nebulizers

- 24-40%(0.24-0.40) FiO_2

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO_2

47:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

48:

Blood pressure is considered labile when it is extremely variable and difficult to control.

49:

Direct current cardioversion is recommended for patients with arrhythmia and hemodynamic compromise (January et al., Circulation 2014, 130: 2071-104).

50:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

51:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

52:

Hypertension is defined as a systolic blood pressure (SBP) of greater than 130 mmHg or a diastolic blood pressure (DBP) of 80 mmHg or greater. Recommendations for initial management may include a conservative approach with diet and lifestyle modification with or without medications. The goal of antihypertensive therapy is to reduce morbidity and mortality. In non-urgent situations, blood pressure (BP) control is achieved in the outpatient setting. Goals for target BP level should be individualized, but a SBP less than 130 mmHg and a DBP less than 80 mmHg are desirable. Populations at high risk include patients with the following conditions (Whelton et al., Hypertension 2018, 71: 1269-324):

- Clinical cardiovascular disease or 10-year atherosclerotic cardiovascular disease (ASCVD) risk 10%(0.1)
- Heart failure
- Stable ischemic heart disease
- Chronic kidney disease
- Chronic kidney disease after renal transplantation

53:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

54:

Management of patients admitted with a heart failure (HF) exacerbation may include supplemental oxygen, diuretic, guideline-directed medical therapy (GDMT), which includes angiotensin-converting enzyme inhibitor (ACEi) or angiotensin receptor blocker (ARB) or angiotensin receptor neprilysin inhibitor (ARNi), beta-blocker, mineralocorticoid receptor antagonists (MRA), and sodium-glucose cotransporter 2 inhibitors (SGLT2i), cardiac monitoring, and deep vein thrombosis (DVT) prophylaxis (Maddox et al., J Am Coll Cardiol 2021, 77: 772-810).

55:

Bilateral pulmonary infiltrates seen on chest x-ray are consistent with the diagnosis of left-sided heart failure.

56:

In patients hospitalized with acute heart failure (HF) due to an increased risk of atrial and ventricular arrhythmias, continuous cardiac monitoring is recommended. Monitoring may be discontinued when there is no evidence of a hemodynamically significant arrhythmia and the symptoms of acute HF have improved (Sandau et al., Circulation 2017, 136: e273-e344).

57:

Intravenous (IV) loop diuretics (e.g., furosemide, torsemide, bumetanide, ethacrynic acid) are administered to relieve the symptoms of dyspnea and congestion without excessively reducing intravascular volume. Due to their relatively short half-life, diuretic effectiveness can be enhanced by continuous administration or multiple boluses daily. Careful monitoring of daily weights, orthostatic vital signs, intake and output, electrolytes, and renal function are key components of diuretic therapy. If a patient does not initially respond to IV diuretics, other options may be considered, including increasing the dose to ensure adequate drug levels reach the kidneys and adding a second diuretic, typically a thiazide, to the loop diuretic in order to improve diuretic responsiveness (Heidenreich et al., Circulation 2022, 145: e895-e1032; Maddox et al., J Am Coll Cardiol 2021, 77: 772-810; Rosendorff et al., Circulation 2015).

58:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

59:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

60:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

61:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

62:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

63:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

64:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

65:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

66:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

67:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

68:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit

- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

69:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

70:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

71:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

72:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

73:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

74:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

75:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

76:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

77:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

78:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

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Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.